

An Extension of the Elo Rating System with Covariates: A Brier Score Analysis

by

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Abstract

This thesis evaluates whether incorporating covariates into paired-comparison rating systems improves predictive accuracy relative to traditional approaches such as the Elo rating system, using Brier score as the evaluation metric. Standard rating systems estimate player strength using only match outcomes, potentially overlooking contextual factors that influence performance.

To address this limitation, this study extends these models by integrating player age differences into the rating framework. Using a large dataset of chess games as the primary case study, predictive performance is assessed using the Brier score and compared against established benchmarks, including classical Elo ratings and official FIDE ratings.

The analysis is further extended to a sports setting to evaluate whether the effects of covariate inclusion generalize beyond chess. Overall, this work contributes to a deeper understanding of how contextual information can be incorporated into probabilistic rating systems and their potential implications for predictive modeling in competitive environments.

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1 Introduction

1.1 What is a rating system?

”A rating system therefore attempts to evaluate all the performances of an individual or a team on some sort of scale, so that at any given time the competitors may be listed in the probable order of their strength. Furthermore, a proper rating system should go a step beyond mere ranking and should provide some estimate of the relative strengths of the competitors, however strength may be defined.” [8]

A rating system is a mathematical framework used to estimate and compare the relative strength of competitors based on observed outcomes. These rating systems are traditionally used in chess due to ratings being based on opponents, so a 1 versus 1 game streamlines this process. Chess has relatively low external randomness compared to many sports, this makes the rating scores more reliable and predictable.

These systems are used in many competitive settings, including board games such as chess, video games, pocket billiards, and even professional sports. Some examples of the different types of rating systems used in chess include:

- The German Chess Federation (Deutscher Schachbund, DSB) uses the Deutsche Wertungszahl system [21]
- The United States Chess Federation (USCF) uses a rating system developed by Mark Glickman [16]
- The International Chess Federation (FIDE) uses a rating system based on the Elo rating system [10]
- Chess.com uses the Glicko-1 rating system [4]
- Lichess.org uses the Glicko-2 rating system [20]

This thesis reviews several rating systems, including Elo, Glicko-1, and Glicko-2, along with things like strengths, weaknesses, and some of the mathematical and statistical ideas behind them.

1.2 Motivation

Rating systems can be used to answer questions about player skill numerically, instead of trying to estimate who beats whom, which inherently introduces bias. These numbers, or "ratings," can be used to create fair seedings in tournaments, create a standardized ranking of players, ensure fair matchmaking, predict match outcomes, and help measure skill growth over time [8].

Rating systems are also used for tournament sectioning and prize eligibility, meaning you must have a certain rating to compete for certain prizes. Ratings could also be used as a measure of qualification for tournaments or events, with some chess federations even using ratings as a means to measure qualifications for titles[12].

1.3 Problem

Traditional rating systems, such as the Elo model, estimate player strength using only match outcomes and relative ratings. These models are effective, but they assume that all relevant information about player performance is through past results alone. In practice, a player's performance may also be influenced by contextual factors such as age, fatigue, strategy mismatches, traveling mishaps, or other external conditions.

This leads to the central research question of this study: Can incorporating such contextual variables improve the predictive accuracy of rating systems? More specifically, this thesis investigates whether the age difference between players has a measurable and generalizable effect on match outcomes.

While it is evident that professional athletes do not perform as well as they age beyond their peak, does this affect skill in a way that these factors could be generalized to the whole population and used in systems that measure skill for the entire population? This study addresses that gap by formally incorporating age difference into the Elo framework and evaluating its impact on predictive performance.

1.4 Contribution

This study extends the classical Elo rating system by incorporating age difference as a covariate within the expected score function. This model is evaluated against the traditional Elo system and benchmark ratings using the Brier score as a measure of predictive accuracy.

Age difference was chosen as a covariate because it is readily available, interpretable, and plausibly related to performance through factors such as experience, cognitive development, and physical or mental endurance.

By comparing predictive performance across models and datasets (chess and professional tennis), this thesis assesses whether incorporating covariates into the rating system leads to meaningful improvements, and whether such improvements generalize across other competitive spaces.

2 Background

2.1 Elo Rating System

Arpad Elo was a Hungarian-American physics professor at Marquette University, known for creating the Elo Rating System, which is commonly used in rating systems for two-player games, particularly chess [28].

Two major advantages of the Elo system are the statistical foundation, and the fact that this system rewards players who exceed expectations and punishes players who underperform [8]. The system replaced by the Elo system was the Harkness rating system, which rated players based on the quality of their competition at individual tournaments. "Dissatisfied by the apparently arbitrary nature of this approach, Elo purposefully designed his system to be more objective and mathematical. The USCF adopted it at its 1960 meeting in Saint Louis, and FIDE followed suit a decade later in 1970, making the "Elo system" the basis for awarding titles, grading tournaments and rating chess players worldwide." [28]

The assumptions necessary for the Elo rating system are:

- Rating is a single, fixed number
- A player's performance will vary randomly around their true skill
- The difference in ratings is the only variable for win probability
- Each match is independent
- Transitivity of skill

Harkness	Elo
Event- based	Game-by-Game
Uses Tournament Strength (median or adjusted average of players)	Opponent's exact rating
Rating = tournament strength $\pm$ adjustment for results	Rating changes based on expected vs actual result
Adjustments are linear	Adjustments are probabilistic (logistic curve)
Less accurate at small events	Handles any number of games reliably
Ratings Updated after events	Ratings updated after each game
Can be skewed by who is present at tournaments	Does not matter who is present at each tournament

Table 1: Adapted from [8, 27]

The expected score for Player A is:

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}}$$

E_A = Expected Score for Player A R_B = Current rating of Player B R_A = Current rating of Player A
The scale factor is 400, which controls the spread of the logistic curve The base of 10 in this formula represents 10 times odds difference. As an example, if Player A was 1800 and Player B was 1400, their rating difference would be 400. For every 400 of rating point difference, the higher-rated player is 10 times more likely to beat their opponent.

The rating update formula for Player A is:

$$R'_A = R_A + K(S_A - E_A)$$

R'_A = Updated rating of Player A R_A = Current rating of Player A K = a constant that determines how rapidly a player's rating changes after a match $S_A \in \{0, 0.5, 1\}$, and represents the actual outcome: loss, draw, win E_A = Expected score for Player A

Similarly, for Player B,

Expected Score:

$$E_B = 1 - E_A$$

Rating Update Formula:

$$R'_B = R_B + K(S_B - E_B)$$

Actual outcome:

$$S_B = 1 - S_A$$

2.2 Brier Score

To evaluate how well each system predicts match outcomes, the Brier Score is used as a measure of probabilistic prediction accuracy.

Let f_t be the predicted probability of winning for match t , and let $o_t \in \{1, 0.5, 0\}$ represent the observed outcome, and N the number of observed matches. Then the Brier Score is given by:

$$\frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2$$

This metric computes the difference between the predicted probability and actual outcome, squares the result, and finds the mean over all games. Smaller differences between predicted probabilities and actual outcomes produce lower Brier scores, indicating a better predictive model.

2.3 Other Rating Systems

There are many rating systems that are being used for all sorts of applications. This discussion focuses on systems that extend or adapt the Elo framework, particularly those used by online chess platforms

and multiplayer video games. Mark Glickman created the Glicko and Glicko-2 systems that built upon Elo's framework; his systems are being used today by online chess entities [4],[20]. TrueSkill and Trueskill2 were created by Microsoft research teams specifically for addressing rating systems in online multiplayer games [17],[22].

"The problem with the Elo system that the Glicko system addresses has to do with the reliability of a player's rating. Suppose two players, both rated 1700, played a tournament game with the first player defeating the second. Under the US Chess Federation's version of the Elo system, the first player would gain 16 rating points and the second player would lose 16 points. But suppose that the first player had just returned to tournament play after many years, while the second player plays every weekend. In this situation, the first player's rating of 1700 is not a very reliable measure of his strength, while the second player's rating of 1700 is much more trustworthy. My intuition tells me that (1) the first player's rating should increase by a large amount (more than 16 points) because his rating of 1700 is not believable in the first place, and that defeating a player with a fairly precise rating of 1700 is reasonable evidence that his strength is probably much higher than 1700, and (2) the second player's rating should decrease by a small amount (less than 16 points) because his rating is already precisely measured to be near 1700, and that he loses to a player whose rating cannot be trusted, so that very little information about his own playing strength has been learned." [15]

Glicko-2	Elo
Ratings Deviation (RD)	Rating update does not change based on time since last game
Complex, requires computer algorithms	Simple, but can be made easier with computer algorithms
Volatility σ (Consistency)	Does not account for consistency of games
Common in Online Platforms (Chess.com & Lichess)	Common in traditional Sports Applications
[10]	

Table 2: Adapted from [8, 12]

TrueSkill and TrueSkill2 are rating systems designed by a Microsoft research team for addressing challenges in using a rating system in online multiplayer games, specifically [17]:

- Game outcomes often refer to a team of players yet a skill rating for individual players is needed for future matchmaking.

- More than two players or teams compete such that the game outcome is a permutation of teams or players rather than just a winner and loser.)

TrueSkill2	Elo
Factors in Uncertainty about a player's rating	Does not account for uncertainty of a player's rating
Uses Average Skill (μ) of a team	Only uses Player Rating
Converges to "true skill" quickly	Converges to "true skill" but slower than TrueSkill2 comparatively
Designed specifically for team games, and free-for-all scenarios	Designed for 1-on-1 games such as chess
Takes into account match statistics (kills, deaths, quitting the game, etc.)	Only uses Player Rating
Can identify "Smurfs" or players who are on an account which is below their skill level	Only uses Player Rating

Table 3: Adapted from [8, 22]

3 Data and Methods

3.1 Assumptions

The additional assumptions necessary for a system using age as a covariate are:

- The effect of age difference is locally symmetric and constant across the observed age range; The difference in skill between a player who is 40 playing a 35-year-old is the same as between a 35-year-old playing a 30-year-old. This is similar to stationary increments in probability theory.
- Stationary age difference effect; The age effect is the same throughout the population, and does not change according to other factors. That is, the age difference effect used in one game has the same effect as it did on all other games in the population.

3.2 Logistic Regression

Logistic regression is a machine learning algorithm that predicts a discrete or categorical outcome. It estimates the probability of a dependent variable as a function of independent variables, using the logistic function to constrain outputs to the interval [0,1]. The estimated probabilities can then be

interpreted as the likelihood of an outcome, such as one player winning a match. Logistic regression is preferred over linear regression in this context because it constrains predicted probabilities to the interval [0,1] and is specifically designed to model binary outcome variables[19].

Logistic regression was used to estimate the marginal effect of age difference on match outcome probability (code shown in the appendix). The estimated age coefficient α_1 was incorporated as an additive adjustment to the Elo rating difference, treating the coefficient as an approximate Elo-equivalent effect size for predictive comparison purposes. The caveat of using regression like this, which is better suited compared to linear regression, is that the analysis was restricted to games with decisive results to maintain consistency with the binary structure of the model. This restriction excluded drawn games from the analysis, reducing the dataset for chess games from 38,936 to 24,749.

The formula used in this thesis is

$$P(Y_i = 1 | X_i) = \frac{1}{1 + e^{-(\alpha_0 + \alpha_1 A_i + \alpha_2 R_i)}}$$

or equivalently,

$$\log \left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)} \right) = \alpha_0 + \alpha_1(A_i) + \alpha_2(R_i)$$

Where for each game i ,

- Y_i = Binary match result
- $X_i = (A_i, R_i)$ = Set of explanatory variables(covariates)
- A_i = Age Difference, Player 2 age minus Player 1 age
- R_i = Rating Difference, Player 2 rating minus Player 1 rating
- α_0 = log-odds that Player 1 wins when the players are equally rated and equally aged.
- α_1 = Change in the log-odds of winning associated with a one-unit increase in age difference, holding rating difference constant
- α_2 = Change in the log-odds of winning associated with a one-unit increase in rating difference

3.3 Elo with Age Difference

The age difference covariate is incorporated as follows:

$$E_a = \frac{1}{1 + 10^{\frac{(R_b - R_a) + \alpha_1(\text{age}_b - \text{age}_a)}{400}}}$$

The inclusion of age difference allows the model to adjust predicted outcomes based on a measurable contextual factor. The parameter α_1 determines the direction and magnitude of the effect. If $\alpha_1 > 0$, older players are favored; if $\alpha_1 < 0$, then younger players are favored.

Age difference enters as an additive term in the log-odds of the logistic win probability, where the age-difference term modifies the effective rating gap between players, adding a source of information when predicting match outcomes.

3.4 Data

Data was obtained from the Fédération Internationale des Échecs (FIDE), using their monthly rating snapshots as initial values for the chess systems. A loop was created in the code to assess how well the monthly snapshot predicts match outcomes, which was used as a baseline to evaluate how well each of the chess rating systems performs. Notably, FIDE uses a dynamic K-coefficient (K=40 new to the list until 30 games, K = 20 rating < 2400, K = 10 rating reaches 2400 and remains, even if rating drops below 2400, K=40 all < 18 years old, with rating < 2400). FIDE was an amazing resource to use, as players must register to play as a FIDE competitor, therefore the high-level tournaments being used in the match outcomes should well represent the data [11].

Match-by-match data was retrieved from The Week in Chess, with 205 PGN files covering 205 tournaments, highlighting some of the more important events. This dataset contains 39,000 games, allowing for broader conclusions to be drawn [5].

The tennis data was retrieved from Jeff Sackmann's Association of Tennis Professionals (ATP) repository on GitHub. Data from ATP matches was taken from 2000 to 2024. This provided match-by-match data (~70,000 games), and statistics from each game, which were used to evaluate whether additional variables serve as useful indicators of player rating [25].

4 Results

By running these rating systems through the match-by-match data, win probability and actual outcome were computed for each match. By comparing the two using the Brier Score metric, a comparison of how well each rating system predicted outcomes can be evaluated.

4.1 Top 20 Chess Players

Before implementing the rating systems on the full dataset, a smaller subset consisting of the top twenty chess players was analyzed to validate the code and ensure that the rating update procedures were functioning correctly. Using a smaller subset allowed debugging and verification of the algorithms before scaling the analysis to the full dataset of approximately 39,000 games.

This subset was constructed using the FIDE monthly rating snapshot from February 2025, the earliest rating list used in this study. The top twenty players included in the subset were:

- Abdusattorov, Nodirbek
- Anand, Viswanathan
- Aronian, Levon
- Carlsen, Magnus
- Caruana, Fabiano
- Dominguez Perez, Leinier
- Duda, Jan-Krzysztof
- Erigaisi, Arjun
- Firouzja, Alireza
- Giri, Anish
- Gukesh D
- Keymer, Vincent

- Le, Quang Liem
- Mamedyarov, Shakhriyar
- Nakamura, Hikaru
- Praggnanandhaa R
- Rapport, Richard
- So, Wesley
- Vachier-Lagrave, Maxime
- Wei, Yi

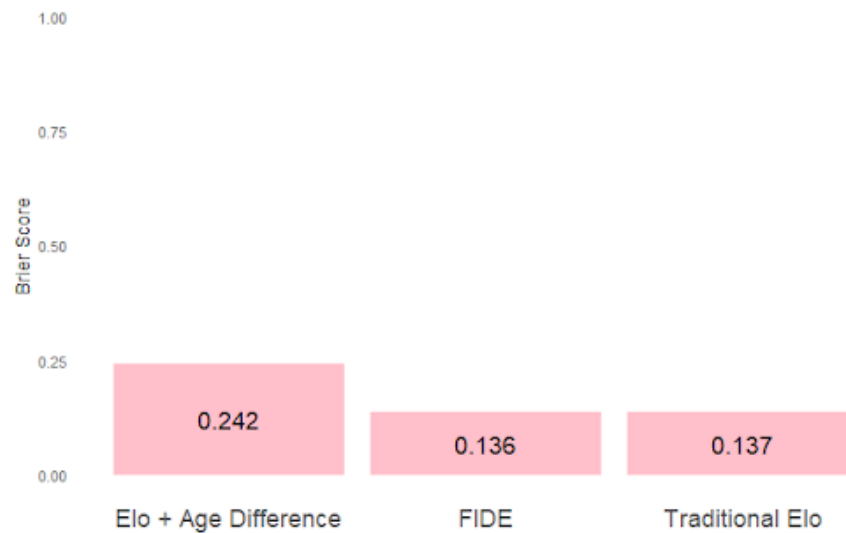


Figure 1: Brier Score for Top 20 Chess Players

The Brier scores indicate that the FIDE snapshots and the traditional Elo system were very similar, with the FIDE benchmark outperforming Elo by approximately 0.001 in Brier score. This likely reflects the use of a variable K-coefficient in FIDE ratings, which allows for the change in ratings to be variable based on contextual factors.

Interestingly, the Elo model with age difference performed worse than both the traditional Elo system and the FIDE benchmark in this subset. The initial hypothesis was that incorporating contextual

factors would help predict the actual outcome of the matches. There could be a few reasons for this result:

- A linear age difference adjustment does not well represent the relationship between age difference and performance
- The effect is too small and inconsistent with the limited sample of players
- Low variance in age between top players

4.2 All Chess Players

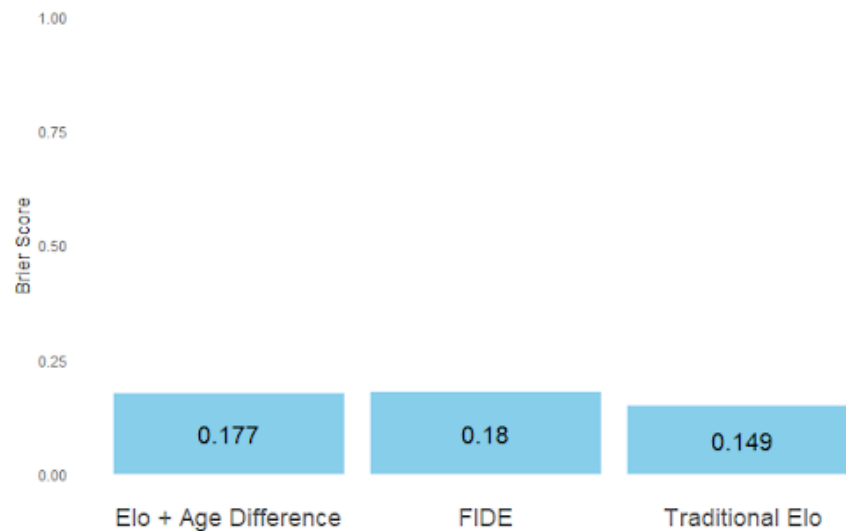


Figure 2: Brier Score for all FIDE recorded chess players

When generalizing this for all chess players, the traditional Elo system outperforms the FIDE snapshots as well as the Elo with age difference system. One strength of the traditional Elo system is that ratings are updated after every match, which would aid in estimating win probability over all players compared to the FIDE monthly snapshot. For all chess players, Elo with age difference improves upon the FIDE snapshot baseline by approximately 0.003 in Brier score, which suggests that age difference may be helpful as a factor when considering win probability in chess. However, it still underperforms the standard Elo system, indicating that the added covariate does not meaningfully enhance prediction when ratings are already updated dynamically.

4.3 ATP Extension



Figure 3: Brier Score for ATP Players

By using the traditional Elo as a baseline, Elo with age difference is a slightly better system when comparing predictive probability. The improvement (approximately 0.004 in Brier score) is modest but consistent, and is similar when compared to the chess Brier scores. This suggests that contextual factors such as age difference may contain modest predictive information in some competitive settings.

5 Limitations

”The measurement of the performance of an individual is always made relative to the performance of his competitors, and both the performance of the player and of his opponents are subject to much the same random fluctuations. The measurement of the rating of an individual might well be compared with the measurement of the position of a cork bobbing up and down on the surface of agitated water, with a yardstick tied to a rope and which is swaying in the wind.” [7]

In essence, Elo ratings attempt to estimate an underlying latent skill from noisy, relative observations. Players may underperform or overperform due to short-term factors such as fatigue, strategy mismatches, traveling mishaps, or just getting unlucky. Also, because Elo is a system built relative to the opponent, these same factors could affect their opponent, where inaccuracies could spread throughout the population of players. Finally, the system can only take in the contextual factors explicitly stated in

the formula. Traditional Elo only takes rating into account, Elo with age difference takes rating and age difference into account. Other systems attempt to address these contextual factors, such as TrueSkill2 implementing match statistics into their algorithms. But, no model fully captures all sources of variation in performance. In the end, Elo and related systems are really just an approximation of skill, not a definitive measure, since they depend on noisy results, modeling assumptions, and the structure of the system itself.

6 Conclusion

This thesis set out to examine whether incorporating contextual covariates into the Elo rating system can improve the predictive performance over the traditional Elo rating system. By extending the standard framework to include factors like age difference, and evaluating the performance using the Brier Score, this study provides empirical evidence on the value and limits of enriching rating models with additional information.

The results suggest that incorporating covariates such as age difference does not uniformly improve predictive accuracy. While modest gains are observed in larger datasets, the effect is inconsistent and sensitive to model specification. This suggests that contextual variables may provide incremental predictive value, but are not sufficient on their own to substantially outperform well-calibrated baseline systems such as Elo.

Extending this idea beyond chess and finding consistent results implies potential applicability across competitive domains, although the magnitude and consistency of improvement may vary depending on the structure of the sport.

At the same time, the findings further prove the idea that rating systems are inherently imperfect representations of skill. As discussed in the limitations, performance is noisy, context-dependent, and only partially observable through match outcomes. Even with contextual factors, models can not fully account for all factors that influence player skill at one point in time. Therefore, improvements of this system should be seen as incremental refinements rather than fundamental changes.

Rating systems appear to already capture most of the predictive signal through ratings alone, with contextual covariates providing only marginal gains unless modeled with greater structural sophistication. Simple covariate augmentation is insufficient; meaningful improvements likely require richer

structural models (e.g., nonlinear effects, interaction terms, or uncertainty-aware systems like Glicko).

7 Future Work

Future work could explore several extensions to improve both the predictive accuracy as well as the structural realism of the models used in this study.

First, the assumption that age difference has a linear effect could be improved in a future model. In reality, the relationship between age and performance is likely nonlinear (e.g., peak performance followed by decline), meaning the linear specification used here is a simplifying approximation. A future model could optimize the age difference covariate so it more effectively represents real-life age differences.

Second, implementation of these rating systems could possibly be streamlined using the `PlayerRatings` package in R, instead of coding a new loop for every rating system. Although a detailed exploration of the `PlayerRatings` package was outside the scope of this study, this package may be helpful for future works in order to increase focus on the formula as well as history, rather than devoting substantial effort to implementation details.

Third, this study focuses on age difference as the sole contextual covariate. Additional factors such as tournament type, travel distance, or recent workload could all affect a player's skill. Additional contextual variables may be especially beneficial in high-variance environments (e.g., sports) compared to lower-variance environments like chess.

Future research could explore more advanced frameworks incorporating uncertainty measures, such as Glicko and Glicko-2, or multiplayer rating systems such as TrueSkill and TrueSkill2. Extending the covariate-based approach within these systems may provide a more flexible and realistic representation of a player's skill over time.

Finally, a weakness of this model was the exclusion of draws due to the logistic regression framework. This could be addressed by including draws using ordinal logistic regression, or there is also an extension of the Elo rating model, the Elo-Davidson model, used especially in sports with a higher number of draws, such as soccer.

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8 Appendix

The full implementation is available at: <https://github.com/JoshuaRSanchez/An-Extension-of-the-Elo-Rating-System-with-Covariates-A-Brier-Score-Analysis-Code.git>

Technologies used: R, RStudio